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LOOMING between shaft looms and a TC2 Jacquard Loom - A Serendipitous Expedition with Live Weaving

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This paper describes an artistic exploration of the potential for more lively engagements with a TC2 digital Jacquard loom. We listened to the loom to enable control of every warp thread in real time, allowing live manipulation of the patterns. This exposed us to the cognitive challenge of designing live for 2 to the power of 1320 warp possibilities. To address this, we made a customised TC2 control software interface, LOOMING, that implements traditional shaft loom logic as a digital, interactive weaving draft. Our tests with two experienced weavers suggest that this facilitates not only a creative space that is workable and exploratory, but also access to traditional weaving knowledge built on the foundation of shaft looms. The interactive weaving draft also offers the versatility to emulate shaft looms with customised shaft numbers and sizes, tie-ups that can be modified on the fly, digitally controlled treadling, and the ability to work with algorithmic logic in a weaving draft. We propose a broader reconsideration of the capability of existing fabrication machines through the semantic space of textile practices, and consider how the 'language' of textile making might be transferable to digital craftsmanship.

Keywords: weaving; weaving draft; semantic space; live coding;

1 Introduction

Discussion around weaving with a computer-controlled loom often invokes early mechanical computers, which appropriated punch cards and automated machinery from the weaving industry, such as the 17th-century Bröselmaschine and 19th-century Jacquard device (Harlizius-Klück, 2017). Such machine looms ingest binary information in order to automate the production of textile patterns, similarly to how computers automate the calculation of algebraical patterns (Lovelace, 1843). Over time, such machine looms supplanted human labour in industrial weaving, just as computers have now supplanted human calculating.

Despite this industrial automation, the human desire to engage directly with both textiles and computation remains. While technological progress has led to the growth of inhumane sweatshops, hand-weaving traditions still survive worldwide (Mamidipudi, 2019). Similarly, with the ongoing automation of software engineering, computer programming has developed its own traditions in immediate expressions, including live coding in the performing arts (Blackwell et al., 2022). Just as weavers make textile patterns by working between weaving drafts and yarn manipulation on the go, live coders make musical, visual or choreographic patterns by writing and manipulating code in live, artistic performance (Blackwell et al., 2022).

In our interdisciplinary collaboration, we are interested in how weaving and live coding can intersect. Both traditions feature discrete, formulaic representations of algorithms, explore patterns through repetition, symmetry and interference, and ground abstract calculation with physical, bodily outcomes, whether weaving garments or live coding music for dancing to.

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The TC2 loom by Digital Weaving Norway has become a key site for research into human-centric creativity with computer-controlled looms. The TC2 is a hybrid human/computer-controlled loom designed for prototyping, where the weft is passed by hand, but where pneumatic control modules allow selection of individual warp threads. This loom is normally operated by first uploading a file in the form of a binary raster image, and then selecting successive warp lifts using a single foot-switch. While this hands-on approach allows significant live intervention by the weaver, changing the lift pattern is still awkward, requiring a new image to be uploaded each time. However, work by Albaugh et al. (2020), Devendorf et al. (2023), and Wu et al. (2024) has demonstrated great potential for developing alternative software interfaces for the TC2, prompting us to investigate the space of programming *liveliness* with the TC2 ourselves.

Driven by this curiosity, we embarked on a journey to reimagine the TC2 in the spirit of live coding. We approached it as a hybrid instrument; a digital system for performing embodied computation (Lin & Witkowski, 2025). We aimed to make the act of weaving akin to live musical improvisation, following the work of Wu et al. (2024) and Albaugh et al. (2020) with a focus on the performative aspect of the live act of weaving. This desire for *liveliness* posed an immediate challenge that emerged from the loom's free control of individual heddles, and the desire to manipulate a pattern live, while weaving. Our TC2 loom has 1320 heddles, which means a row of the digital pattern consists of 2 to the power of 1320 possible combinations – an astronomical number with 398 digits! How could we navigate such a huge space of possibility while retaining its immediacy? This challenge led us to explore traditional textile practices and their semantic space, in particular weaving with shaft looms. We built a 'live weaving' interface – 'LOOMING' – that emulates a traditional shaft loom with direct communication to the TC2, turning it into a real-time, interactive loom with live reprogrammable threading, tie-up and treadling. This allows live improvisation with the configuration of a notional shaft loom, while responding to ongoing material results. We are finding that weaving with 'LOOMING' as an interactive weaving draft allows weavers to explore the creative space of embodied materiality, engage with structural design, and think within the semantic space of the floor loom, thereby joining a long history of weaving traditions.

In this paper, we argue that technological innovations in textiles could benefit from stronger connections to the cognitive and semantic space of existing craft practices by adopting the existing 'languages' of textiles in order to enable designers to create textile-native designs while exploring and improvising live at the loom.

2 A Design Research Journey Driven by Artistic Motivation

We work as an interdisciplinary collective. The first author is a bio-artist and new media artist who focuses primarily on collaboration with non-human others. The second author is a software artist, live coder, and researcher, bridging interests in the algorithmic patterns of weaving, music, and other pattern-based arts and crafts. The third author is a design researcher with a background in textile design, specialised in weaving. The fourth author is a textile designer working with material agency and the unknown. The fifth author comes to weaving from a background in experimental music and an ongoing practice of design research.

Together, we take a Research through Art and Design approach (Frayling, 1994) with a focus on exploration, allowing artistic activities to emerge in the process, revealing and contextualising questions along the way. This article is written as a report on a series of explorations, thought processes, challenges, innovations, and contextualisations that together bring insights to our original point of departure.

3 Weaving as Live Coding

In the book *Live Coding: A User's Manual*, Blackwell et al. (2022) describe *Live Coding* as follows: 'Live coding is about people interacting with the world, and each other, in real time, via code. Live coding is about making software live' (Blackwell et al., 2022). While live coding is best known in performing arts and especially music, live coding research also considers the live relationships between coder, their code, and the materials and/or other artists they are working with through that code, including in cases where an audience is absent McLean et al., 2023.

Live coding, therefore, shifts computer programming from the process of producing programs into ad hoc, lively engagements for programmers, where code is written and manipulated while it runs. Weaving, a practice that shared the same heritage as programming (Lovelace, 1843), is also a fundamentally digital and computational way of making (Lin & Witkowski, 2025) – a physical process where yarns are interlaced to make textiles. The invention of computer-controlled looms, traverses the boundary between digital coding and physical making, allowing binary code to be embodied into a tactile textile, but misses out on a handweaver's live, physical experience of engaging with weaving structures on a handloom. As a result, incorporating weaving into live coding became an intriguing possibility, inspired by the way that robot arms, bodily movements, and musical instruments have already been used for live coding performances (McLean & Sicchio, 2024; Sicchio, 2014; Sicchio & McLean, 2017). Following earlier work with live coding a handmade loom limited to 16 threads (McLean, 2026), we decided to explore this connection on the TC2 loom.

4 The TC2 Loom and its Position in HCI Research

The TC2 is often used in HCI research into weaving due to its hybrid manual and automated nature (Voorwinden et al., 2025). It has been used to explore e-textiles (electronic textiles), woven structures, and digital craft innovations (Voorwinden et al., 2025; Wu et al., 2024). Because the TC2 control software does not offer functions to edit the digital pattern, the designs are often done in image editing software such as Adobe Photoshop or GIMP before being loaded into the TC2 control software (Schlein & Ziek, 2007; Wu et al., 2024). Professional weaving software used in the industry that allows designing the drawdown in the form of weaving drafts is also available, such as NedGraphics (NedGraphics, 2025), ScotWeave (ScotCad Textiles Limited, 2025), Penelope CAD(Informàtica Tèxtil SL., 2026), and PointCarre (Pointcarre Textile Software, 2026). In addition to industrial software, the web-based software AdaCAD combines graphical programming and weave drafts to make e-textiles (and other kinds of textiles) using the TC2 (Devendorf et al., 2023; Friske et al., 2019) and Draft Player with customised loom pedals by Wu et al. (2024), which allows real-time improvisation on the patterns. The workflow for all these options except for the Draft Player is constrained by the TC2 software, in which live modification of the design during the weaving process is not officially supported. However, previous research by Albaugh et al. (2020) and Wu et al. (2024) shows that it is possible to interact with the TC2 loom in real time by communicating via the TCP/IP protocol and bypassing its control software, which led us to explore developing an alternative control interface for the TC2 to enable its potential for *liveliness*.

5 *Liveliness* and its challenge

We followed the experience from Albaugh et al. (2020) and Wu et al. (2024) and identified the communication protocols through listening to the communication with Wireshark (Wireshark Team, 2026). This allowed us to identify and control each warp thread on the TC2 individually. Using this information, we developed a host and client system using Python and the MQTT protocol that would allow direct communication with the TC2 loom, and began to build a customised client interface.

While building the interface for real-time interaction, we encountered the challenge mentioned in the

introduction: the possible combinations in a row of a digital pattern are astronomical, presenting a challenge for making changes in real time. In the early prototype, we explored two directions: pixel-by-pixel modification and algorithmic generative patterns. These two versions each have their own pitfalls, when it comes to liveliness. Pixel-by-pixel modification is time-consuming, as each pixel change requires one mouse click. The algorithmic generative pattern is unintuitive for non-coders, as it requires the weaver to 'compile' the algorithm in order to understand what the outcome might look like. In addition to this, the algorithms are disconnected from the textile qualities and instead treat the warp and weft interlacement as a bi-colour digital display. Instead, we needed a thinking tool that would allow us to speak a textile language, and enable textile design to occur live, while weaving. The challenge and opportunity of such *liveliness* can be described through two interconnected concerns: the freedom of live improvisation, and the temporality of weaving as a kind of choreography.

In terms of choreography and temporality, a high degree of synchronisation emerges between the flow of design information and the embodied movements of the weaver in relation to the loom, when the information and execution are at the same level of complexity. Weaving has a strong connection to music and dance across cultures and countries. For instance, the Balinese Tenun weaving dance (Suparwa et al., 2024) and the Nottingham Lace Scottish dance (Freemantle & Abbott, n.d.). These connections are not just interpretations or embodied representations; there are also examples of weaving songs being used as a notation system for different patterns while warping (qemeljesai) in Paiwan culture from Taiwan (許春美, 2020), and patterns of weaving (Naqshe Khani) while weaving in Persian tapestry in Iran (Aminian, 2019). These examples exhibit a high degree of synchronisation in the temporality of bodily movements guided by the written scores. The temporal synchronisation between a tune, bodily movements of weaving, and the build-up of textile from weaving happens because structurally they resonate with each other: the tune corresponds to the bodily movements that mediate between the physical structures of the loom, which contributes to the speed of fabric build-up. However, for these weaving and music cultures, there is little room for improvisation while weaving, as the song represents an existing fixed pattern and reflects the loom structure.

From the perspective of improvisation, frame looms, shaft looms, and the TC2 provide different degrees of freedom. The freedom of live improvisation decreases as the level of automation increases, which is closely related to the physical design of the looms as previously discussed by Wu et al. (2024). Computer- and Jacquard-controlled looms have promised great flexibility in weaving by enabling warps to be automated individually, making it possible to create any combination of bindings within the loom's physical limitations. However, it requires full pre-planning for the design, which offers limited space for improvisation during weaving. In modern computer-controlled looms, design is done in a computer before being loaded onto the loom, separating the design process from the making and creating a greater latency between design and seeing its physical feedback, as Digital Craftsmanship attempts to disrupt (Andersen et al., 2019; Frankjær & Dalsgaard, 2018; Goveia da Rocha et al., 2021). In a TC2 loom (as conventionally used), the only improvisation option during weaving is the choice of weft yarns. In a frame loom, the weft can easily be partially woven or woven across the whole fabric while the weaver threads the weft through the warps, giving maximum freedom as long as the warps and wefts intertwine. In a shaft loom, the warps are grouped in shafts and tie-up, lifted in groups with different combinations as chosen by the weaver while they weave, allowing freedom of order in the warp groups.

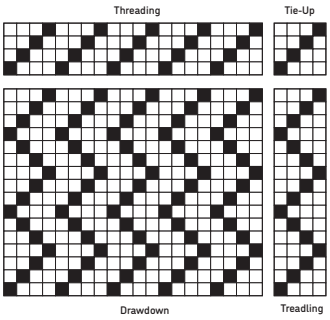
Drawn by the beauty of being able to improvise on-the-go with immediate material feedback from the treadling and lift plans shaft looms, we began to ponder the difference between designing with Photoshop for a TC2 loom versus designing with the table loom and a weaving draft. In terms of cognitive complexity, frame looms and Jacquard looms operate at the granularity of each individual warp-weft interlacement as discrete units. In contrast, shaft looms work in a compressive logic by combining shafts in groups, and design through exploring their relational structures by implementing threading, tie-ups,

and treading, allowing simple changes to be generative of complex and often surprising results.

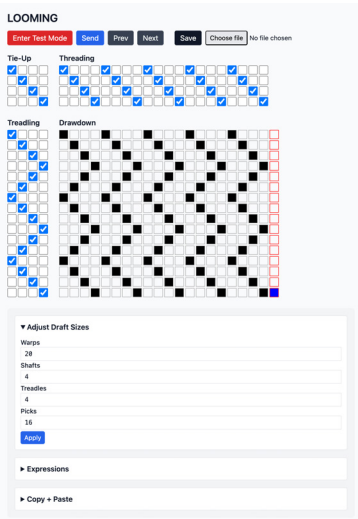
The notation system for shaft looms takes the form of a *weaving draft*, which directly reflects the physical structure of shaft looms. A more abstract mathematical algorithm, on the other hand, uses symbols and logic that require additional calculations to reveal the pattern, demanding an extra layer of cognitive resources. Drawing on the above observations, we oriented our inquiry towards adopting the semantic space of shaft looms through working with a weaving draft that is intrinsically generative. The motivation for adopting the weaving draft was reinforced by a practical challenge: the second author lives in a different country without access to the TC2 loom, but only to a shaft loom. If we want to enable an asynchronous performative weave that could happen in two different countries, bridging the semantic space between a shaft loom and the TC2 becomes necessary.

6 Adopting a Weaving Draft

A weaving draft is a cognitive tool for weavers, facilitating exploration within the physical limitations of shaft looms. It is designed with structures that correspond to different parts of a shaft loom: each row of the threading corresponds to one shaft, each column of tie-up corresponds to a treadle, and each row of the treadling decides which treadles get activated for each weft pick. Threading, tie-up, and treadling together generate warp-weft interlacement, which is represented by drawdown in a weaving draft (Figure 1a).



(a) *A traditional weaving draft.*



(b) *LOOMING interface.*



(c) *LOOMING interface while tested by two people. Can be used by one person when the screen is placed closer to the loom.*

Although a weaving draft maps to the physical parts and actions of a frame loom, it is important to recognise that it doesn't fully represent the weave itself. The drawdown may resemble the appearance of the surface of the cloth, but not always, as the behaviour of threads interlacing in three dimensional space have their own behaviour, which can be difficult to predict. A naive weaver may be surprised to find their woven cloth splitting into layers, tensions creating diagonals, and other surprising behaviours which may only reveal themselves when warp tension required during weaving is reduced. Even an experienced weaver will delay judgements of their weaving draft design until they have actually woven it. From this perspective, the possibilities of interacting with a draft live, while weaving it, are exciting.

We designed LOOMING (Figure 1b) with a mirrored form of a weaving draft (Figure 1a) following the screen logic of reading from the top-left. It is both a virtual shaft loom and an interactive cognitive tool for designing weaving structures. LOOMING allows customisation of the number of warps, shafts, treadles, and picks, to emulate any possible shaft loom setup. Threading, tie-up, and treadling can be

changed on the go, while the drawdown is generated accordingly. We set the maximum number of threads to 1200 warps with a 60-end selvedge on each side, mapping to the full 1320 warps of our TC2 loom. When customised warps are less than 1200, LOOMING automatically repeats the pattern until it reaches 1200 pixels in width and creates a digital pattern of 1320 pixels wide with selvedges. LOOMING sends the digital pattern row by row to TC2, starting from the bottom of the weaving draft, progressing upward until reaching the top. Once the top row has been reached, it starts again with the bottom row, sending another iteration of the digital pattern.

7 LOOMING in Action

LOOMING was developed through use, and in the following, we share the weaving journeys taken with the software by two of the authors.

7.1 The third author's weaving journey.

The third author participated in the discussion of LOOMING's interface design, helping us unpack the cognitive differences between designing with traditional weave logic and weaving on the TC2. They describe how traditional weave logic works in 'blocks', as slow and fixed due to the threading being determined before weaving. Meanwhile, the TC2 provides immediacy and variability, but is hindered by having to design ahead of weaving. With this in mind, she began using LOOMING with traditional weaving logic and made samples that unexpectedly possessed a tea towel aesthetic (Figure 2). Their weaving journey is shown in Figure 3. Their first trials of the interface (sessions 2 and 3) simply explored the use of 'blocks' using passive yarns, defined areas where differing patterns alternate across the width of the warp in visually contrasting 'block' forms. This design technique is lifted directly from traditional table loom usage, where the width and repetition of these blocks are pre-planned into the making of the warp and cemented in the threading pattern of shaft groups. Of course, the TC2 does not function in this way (because there are no shafts and every end can be lifted individually), but the simplification of using the loom in this way allowed for a realignment of their knowledge and provided an interesting design space where the woven sample might look both analogue and digital. In session 3, they took advantage of threading being changeable through LOOMING and explored the drift of a block draft that was previously almost impossible for shaft looms. The details of the changes can be seen in Figure 4, which shows that the changes between the modification steps in LOOMING were rather minimal. In session 4, they explore another element that was not changeable for table looms - the tie-up - through single point changes. In session 5, hybrid techniques that take advantage of the flexibility of TC2 loom with the shaft loom weaving logic have been woven into the sample.

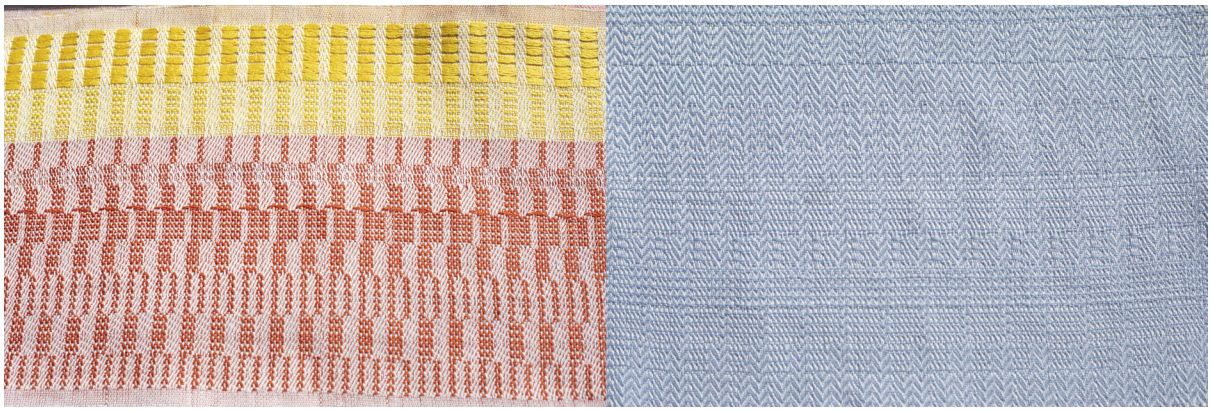


Figure 2: *The third author's sample from weaving session 2 (left) and weaving session 5 (right).*

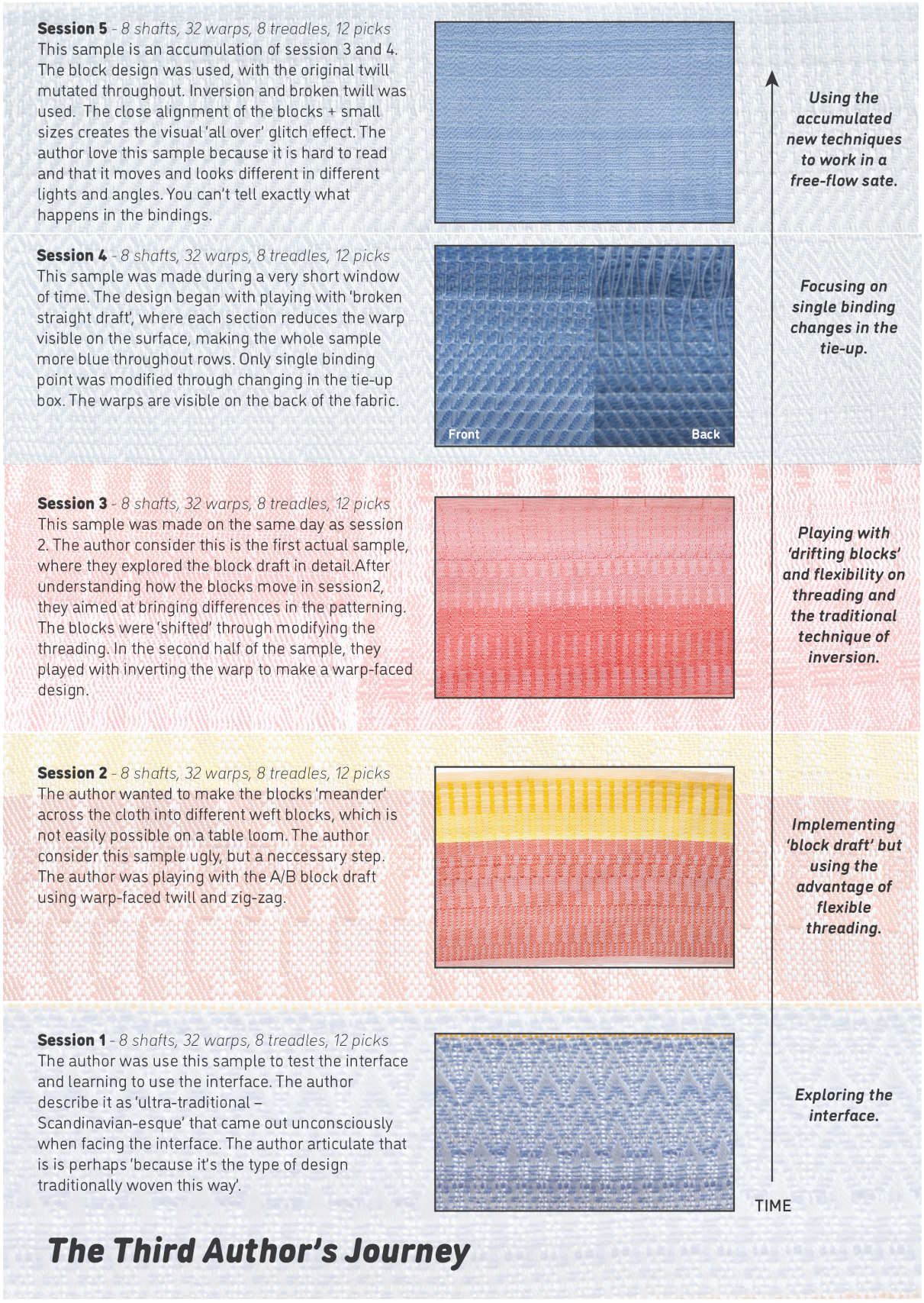


Figure 3: *The third author's weaving journey, where the journey starts from the bottom of the image and works upwards throughout time.*

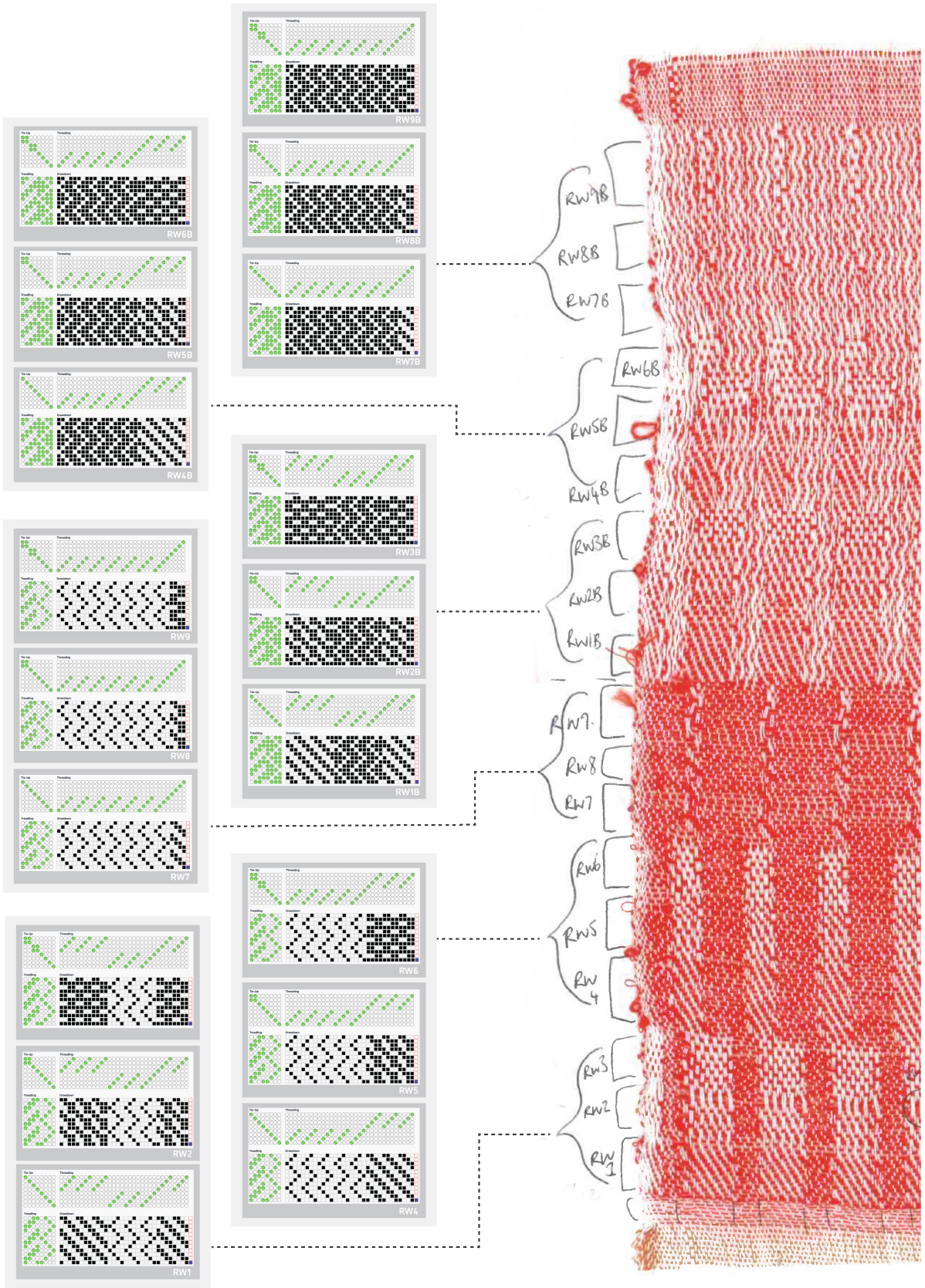


Figure 4: *Detail process of the third author's session 3, showing each of the weaving draft settings used in correspondence to the textile. The fabric builds up from the bottom to the top.*

7.2 The fourth author's weaving journey.

Figure 6 shows the weaving journey of the fourth author while weaving the textile sample in Figure 5. They are accustomed to working with digital Jacquard weaving through designing in Photoshop, having had only one year of shaft loom classes. They used LOOMING to explore an over-twisted wool yarn. Their journey began by exploring LOOMING's functionality for some rows before starting session 1. It helped them build an understanding of the interaction between the interface, the logic of weaving drafts, and the material reactions on the loom as the warps and wefts intertwine. We see from session 1 that the weaving draft logic helps them to quickly prototype a balanced fabric through inversion. In session 2, they began to implement their experience with floats from a digital pattern to the logic of the shaft loom. In session 3, the weaving draft format prompted the fourth author to look into existing bindings from weaving literature, connecting the Jacquard loom knowledge with shaft loom knowledge. In session 4, the flexibility of being able to change the number of shafts, warps, and treadles allowed the fourth author to scale up their design from 8 shafts to 24 shafts, which is impossible for a physical shaft loom. We see the cognitive progression of the fourth author from a digital Jacquard loom (session 2) and a shaft loom (sessions 1 and 3) converge in session 4, where both scaling (a digital Jacquard loom function) and the generative nature of weaving drafts were at work. The immediate feedback between the digital pattern and the materiality also benefits the fourth author in their design decisions, as they express that it was much more intuitive than designing in Photoshop.



Figure 5: *The fourth author's first textile sample woven using LOOMING.*

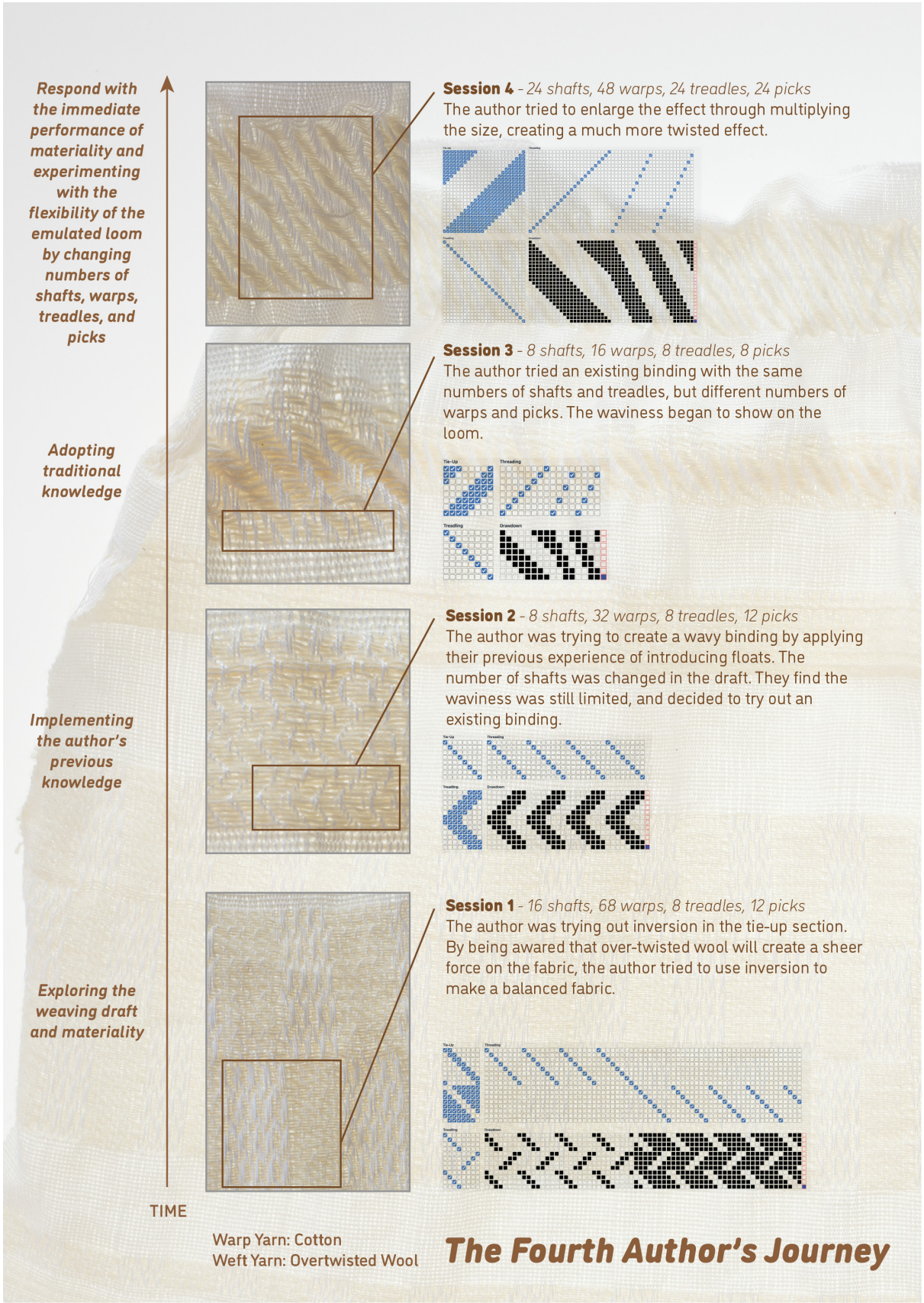


Figure 6: *The fourth author's weaving journey, showing the weaving process from the bottom to the top with corresponding weaving drafts used for each session.*

8 Discoveries

As we wove, we found that LOOMING provides a design space with shaft-loom and weaving-draft semantics, flexibility on emulating any physical set-ups of shaft looms, and an immediate pattern-material feedback loop that was only previously available for hand looms. We see evidence of this throughout our observations and specifically observed the following two changes unfold during our trials.

8.1 Immediate Response and the Shifting of Attention

LOOMING made the interplay between the weaving semantics and the material apparent. This was made possible through multiple factors: 1) the interactive interface allows immediate changes on the TC2, 2) the weaving draft semantics, where one click can make generative changes in a repetitive, rhythmic manner, and 3) the internal repetition function to fill the design over the whole width of TC2 warps, making the textile automatically become a texture. Together, these factors serve as an immediate amplifier for the weaver's action, allowing them to quickly test how bindings react with the yarn they use. In the fourth author's journey, the amplification brings cognitive clues for the yarn's performativity in correlation to the binding (Figure 6 between sessions 2 and 3); in the third author's journey, a single change will be immediately reflected in the woven textile, drawing attention to the details of a binding. Design decisions became more intuitive, and the weavers were able to pay more attention to the bindings rather than the imagery-oriented Jacquard-style weaving designs.

8.2 Connecting to the Traditional Weaving Semantics

The shift of attention from imagery to bindings prompted us to look deeper. The third author explicitly expressed that they had been taught that digital Jacquard-style TC2 looms were made for weaving images, with typical bindings, such as varieties of satins and twills. Working with LOOMING awakened their pre-existing knowledge of designing in blocks, point drafts, and other traditional weaving semantics. Having less experience with shaft looms, the fourth author also revisited shaft loom knowledge to open up a different design space. The textile outcomes possess tea towel-like aesthetics, demonstrating optical effects and textural details that are less often seen in Jacquard-loom-made fabrics.

As a reflection, we consider these outcomes through the angle of 4E cognition (Bruin et al., 2018), where cognition is considered *embodied, embedded, extended, and enactive* (Bruin et al., 2018) through the tools and environment with which the human body engages. Weaving, from this perspective, is embodied in the weaver's body, embedded in the environment where it happens, extended to the tools and materials used, and enactive through the interactions (Hutchins, 1995; Kirsh, 2013; Suchman, 2007; Varela et al., 1991, 2017) between the weaver, the machine, and materials. This entanglement also reciprocally influences thought, where empirical proofs (Varela et al., 1991, 2017) show that tools shape human cognition. This bidirectional interaction is described by Kirsh (2013) such that when designers design tools, they create an enactive landscape that influences thinking. Harlizius-Klück and McLean (2021) described the difference between working with a hand loom and a Jacquard loom as follows: 'A weaver at a handloom can shift naturally between thinking in two-dimensional grids, and working in the three-dimensional textile and the features of the material in front of her. A weaver at a software-controlled machine loom today designs images on a computer screen.' By implementing an interactive weaving draft with a TC2, LOOMING provides a hybrid semantic space connecting the traditional weaving knowledge with the digital screen-based design space. This connection enables transformative creativity (Boden, 2004) to occur, as it speaks the language of a shaft loom as embodied in a digital Jacquard loom, creating different affordances to the Jacquard weaving design.

We believe that such hybridity strengthens the continuum from traditional weaving to digital fabrication, highlighting how digital technologies can bring transformation to the traditional craft practices within the crafts' semantic space rather than practising craft in the digital semantic space. It emphasises the

'craft' of Digital Craftsmanship rather than the digital.

9 Conclusion

In live coding interactions, computer expressions that allow changes to be immediately perceived by the coder play a key role, and it is similarly so for *live weaving*. In this exploration, we chose to see weaving as a performative action, and the textile outcome as a performative artefact. The performative artefact then becomes the sedimentation of the performance (Lin et al., 2024), the medium for the audience to understand, and the visualisation of the happenings between the human weaver, the yarns, the machine, and the pattern/code. The enabling of immediately noticeable changes through real-time communication between the weaver and the loom allowed us to bridge the shaft loom semantics with the digital Jacquard loom hardware. We present LOOMING as a real-time interface for the TC2, borrowing from shaft loom semantics, enabling the *liveliness* of weaving and leading to the serendipitous discovery that such emulation strengthens the connection to traditional weaving knowledge, and thereby making the traditional intrinsic within the digital embodiment. Through the weaving sessions with LOOMING, we found that the shaft loom aesthetics emerge naturally, and as a result, we advocate that traditional weaving knowledge should be revisited. We believe that by studying existing semantic spaces of crafts, we have the opportunity to reconsider synthesising the connections between these semantic spaces and the practice of Digital Craftsmanship.

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